

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06 24	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	P. Harthik	Reg. No.:	192524222
Section / Batch:		Date:	18/7/26

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I P. Karthik - 192524222 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Karthik

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)						
Q2 (30 Marks)						
Q3 (30 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO-2 AT-1

Name: P. Karthik

Reg.no: 192524222

Code: CSA 0624

Course: DAA

1) Selection sort:

[64, 25, 12, 22, 11]

Given array

64	25	12	22	11
----	----	----	----	----

Pass 1: smallest element $\rightarrow$ 11

Swap 64 and 11

11	25	12	22	64
----	----	----	----	----

Comparisons = 4

Pass 2: remaining array

25	12	22	64
----	----	----	----

small = 12

Swap 25 and 12

11	12	25	22	64
----	----	----	----	----

Comparisons = 3

Pass 3:

25	22	64
----	----	----

small = 22

11	12	22	25	64
----	----	----	----	----

Comparison = 2

Pass 4 :

25 64

small = 25

No need to swap

comparision = 1

Total comparisons = 10

Final sorted array

11 12 22 25 64

Time complexity:

Case	Complexity
Best	$O(n^2)$
Average	$O(n^2)$
Worst	$O(n^2)$

2) Closest pair - Brute force Analysis.

Given,

(2,3), (12,30), (40,50), (5,1), (12,10), (3,4)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance calculation:

$P_1 = (2,3)$ & $P_2 = (12,30)$

$$= \sqrt{10^2 + 27^2}$$

$$= \sqrt{829}$$

$$= 28.79$$

$$P_1 \& P_3 \quad P_1 = (2, 3) \quad P_3 = (40, 50)$$

$$= \sqrt{38^2 + 47^2}$$

$$= \sqrt{3653}$$

$$= 60.44$$

$$P_1 \& P_4 \quad P_1 = (2, 3) \quad P_4 = (5, 1)$$

$$= \sqrt{9 + 4}$$

$$= \sqrt{13}$$

$$= 3.60$$

$$P_1 \& P_5 \quad P_1 = (2, 3) \quad P_5 = (12, 10)$$

$$= \sqrt{100 + 49}$$

$$= \sqrt{149}$$

$$= 12.20$$

$$P_1 \& P_6 \quad P_1 = (2, 3) \& P_6 = (3, 4)$$

$$= \sqrt{2}$$

$$= 1.41$$

$$P_2 \& P_3 \quad P_2 = (12, 30) \quad P_3 = (40, 50)$$

$$= \sqrt{784 + 400}$$

$$= \sqrt{1184}$$

$$= 34.40$$

$$P_2 \& P_4 \rightarrow P_2 = (12, 30) \quad P_4 = (5, 1)$$

$$= \sqrt{49 + 841}$$

$$= 29.15$$

$$P_2 \& P_5 \rightarrow P_2 = (12, 30) \quad P_5 = (12, 10)$$

$$= \sqrt{400}$$

$$= 20$$

$$P_2 = (12, 30) \quad P_6 = (3, 4)$$

$$= \sqrt{81 + 676}$$

$$= \sqrt{757}$$

$$= 27.31$$

$$P_3 = (40, 50), P_4 = (5, 1)$$

$$= \sqrt{1225 + 2401}$$

$$= \sqrt{3626}$$

$$= 59.03$$

$$P_3 \& P_5 \rightarrow P_3 = (40, 50), P_5 = (12, 10)$$

$$= \sqrt{784 + 1600}$$

$$= \sqrt{2384}$$

$$= 48.83$$

$$P_3 \& P_6 \rightarrow P_3 = (40, 50), P_6 = (3, 4)$$

$$= \sqrt{1369 + 2116}$$

$$= \sqrt{3485}$$

$$= 58.69$$

$$P_4 \& P_5 \rightarrow P_4 = (5, 1), P_5 = (12, 10)$$

$$= \sqrt{130}$$

$$= 11.4$$

$$P_4 \& P_6 \rightarrow P_4 = (5, 1), P_6 = (3, 4)$$

$$= \sqrt{13}$$

$$= 3.61$$

$$P_5 \& P_6 \rightarrow P_5 = (12, 10), P_6 = (3, 4)$$

$$= \sqrt{117}$$

$$= 9.85$$

Closest pair

(2, 3) and (3, 4)

Distance = 1.41

Number of comparisons

There are 6 points

$$n(n-1)/2$$

$$= 6(6-1)/2$$

$$= 15 \text{ comparisons}$$

Time complexity

$$\text{Best} = O(n^2)$$

$$\text{Average} = O(n^2)$$

$$\text{Worst} = O(n^2)$$

Convex Hull - Brute Force Approach:

Given, $(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)$

Hull edges

Checking every possible line pair, the valid boundary edges

$$(0,0) \rightarrow (3,0)$$

$$(3,0) \rightarrow (3,3)$$

$$(3,3) \rightarrow (0,3)$$

$$(0,3) \rightarrow (0,0)$$

Interior points

$$(1,1)$$

$$(2,1)$$

$(2,2)$ are inside the polygon

Total comparisons

No. of points $n=7$

$$\text{possible pairs} \rightarrow 7 \times 6 / 2$$

$$= 21 \text{ pairs}$$

$$\text{Approximate comparisons} \rightarrow 21 \times 5 = 105$$

Time complexity

$$O(n^3)$$

All points lie inside or on the polygon formed by
 $(0,0), (3,0), (3,3), (0,3)$

Hence, the convex hull is correct.

4) Bubble sort - Analytical problem

Given array

45	12	78	34	23
----	----	----	----	----

Pass 1 :

12	45	78	34	23
----	----	----	----	----

12	45	32	78	23
----	----	----	----	----

12	45	32	23	78
----	----	----	----	----

Pass 2 :

12	45	32	23	78
----	----	----	----	----

12	32	45	23	78
----	----	----	----	----

12	32	23	45	78
----	----	----	----	----

Pass 3 :

12	32	23	45	78
----	----	----	----	----

12	23	32	45	78
----	----	----	----	----

Pass 4 :

12	23	32	45	78
----	----	----	----	----

Total comparisons:-

$$4 + 3 + 2 + 1 = 10$$

Total swaps:

$$3 + 2 + 1 = 6$$

Time complexity

Case	Complexity
Best	$O(n)$
Average	$O(n^2)$
Worst	$O(n^2)$

5) Linear search -

Data set $\rightarrow$

93	45	12	67	34	89	10
----	----	----	----	----	----	----

Key $\rightarrow 34$

Compare
 $23 \neq 34$

Comparison 1

$45 \neq 34 \rightarrow$ Comparison 2

$12 \neq 34 \rightarrow$ Comparison 3

$67 \neq 34 \rightarrow$ Comparison 4

$34 = 34 \rightarrow$ Comparison 5

Element found at Index = 4, Position = 5

Total comparisons - 5

Time complexity

Case	Complexity
Best	$O(1)$
Average	$O(n)$
Worst	$O(n)$

Pseudocode:

Linear search (A, key):

for $i = 0$ to $n-1$;

if $A[i] == \text{key}$;

return;

return -1